Choosing Better NLDR Layouts by Evaluating the Model in the High-dimensional Data Space

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Abstract

Nonlinear dimension reduction (NLDR) techniques such as t-SNE and UMAP provide low-dimensional representations of high-dimensional data, capturing complex structures in a concise visual summary. While NLDR has an important role in data analysis, these methods can badly miss the mark by exaggerating existing structures or suggesting spurious patterns. NLDR applies nonlinear transformations to create layouts, and the choice of method and hyperparameters can produce wildly different representations. It is difficult to decide which are reasonable and which are misleading, and we show they can be poor. To help, we have developed a procedure that treats the low-dimensional NLDR layout as a model that can be lifted back into the original high-dimensional space. This enables us to assess goodness of fit by computing the error and visualize the fitted model overlaid on the data using grand tours. These elements allow us to quantitatively compare multiple layouts, determine which representations are acceptable, and examine whether an embedding captures the data structure throughout the entire space, only in certain subspaces, or fits poorly overall. This approach has also proven useful for understanding how different methodological choices affect the resulting embeddings.

Keywords: high-dimensional data, dimension reduction, data visualization, statistical graphics, tour, unsupervised learning

1 Introduction

Nonlinear dimension reduction (NLDR) is popular for making a convenient low-dimensional (k-D) representation of high-dimensional (p-D) data ($k < p$). Recently developed methods include t-distributed stochastic neighbor embedding (tSNE) (Maaten & Hinton 2008), uniform manifold approximation and projection (UMAP) (McInnes et al. 2018), potential of heat-diffusion for affinity-based trajectory embedding (PHATE) algorithm (Moon et al. 2019), large-scale dimensionality reduction using triplets (TriMAP) (Amid & Warmuth 2022), and pairwise controlled manifold approximation (PaCMAP) (Wang et al. 2021).

However, the representation generated can vary dramatically from method to method, choice of hyper-parameter, or even random seed, as illustrated by Figure 1. The choice may result in different procedures being used in the downstream analysis or different inferential conclusions. Various academics have expressed concerns about current practices and procedures for choosing (e.g., Irizarry (2024), Chari & Pachter (2023)). The research described here provides new numerical and visual tools to help choose a suitable layout.

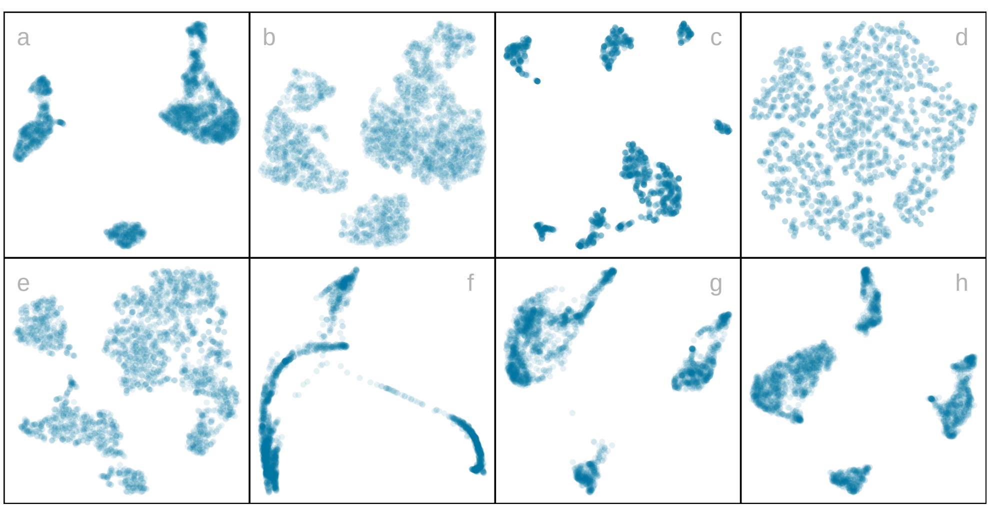

Figure 1: Eight different NLDR representations of the same data, produced by tSNE, UMAP, PHATE, TriMAP and PaCMAP with a variety of hyper-parameter choices. The variety in layouts makes it difficult to choose which best represents the data distribution.

The paper is organized as follows. Section 2 gives more details behind NLDR and high-dimensional data visualization. Sections 3, 4, 5, 6 describes the new methodology, illustrated using a simulated data example. Section 7 extends the approach for prediction of new observations. Section 8 illustrates interactive graphics to diagnose an NLDR fit. A workflow for choosing from a selection of layouts is provided in Section 9. Two applications are in Section 10. Section 11 discusses limitations and future directions. In addition, there is an R package `quollr` which implements the methods.

2 Background literature on dimension reduction

Making useful low-dimensional (k-D) layouts of high-dimensional (p-D) data has historically been achieved using multidimensional scaling (MDS) (Kruskal 1964)]. In MDS, the k-D layout is constructed by minimizing a stress function that compares distances between points in p-D with potential distances between points in k-D. Various formulations of the stress function result in non-metric scaling (Saeed et al. 2018) and isomap (Silva & Tenenbaum 2002). Principal components analysis (PCA) produces the same layout as classical MDS, if interpoint distances are Euclidean. Recent overviews of MDS, PCA and statistically conducting linear and nonlinear dimension reduction can be found in Borg & Groenen (2005), Johnstone & Titterington (2009), Jolliffe (2011), Koch (2013).

The recently developed NLDR methods tSNE, UMAP, PHATE, TriMAP, and PaCMAP, can be considered new approaches tackling the same problem as MDS. The methods tSNE, UMAP, TriMAP, and PaCMAP are similar in that they determine the k-D representation by minimizing the divergence between two inter-point distance distributions. PHATE is different, and is based on a diffusion process spreading to capture geometric shapes that include both global and local structure. (See Coifman et al. (2005) for more details on diffusion processes.)

The array of layouts in Figure 1 illustrate what can result by choosing different methods, tuning with different hyper-parameters, and even the random seed that initiates the computation. Each layout provides a different lens on the interpoint distances in the data. The key structures in the high-dimensional data that we might hypothesize from these 2-D layouts are: (1) highly *separated clusters* (a, b, e, g, h) with the number of clusters ranging from 3-6; (2) *stringy branches* (f), and (3) *barely separated clusters* (c, d). They are at odds with each other, and it begs us to decide which are actually present in the p-D data.

The alternative approach to visualizing the high-dimensional data is to use linear projections, such as PCA. Tours, defined by Asimov (1985), broaden the scope by providing movies of linear projections showing views of the data from all directions. Lee et al. (2021) has a review of tour methods. Many tour algorithms are available in the R package `tourr` (Wickham et al. 2011), and versions enabling better interactivity in `langevitour` (Harrison 2023) and `detourr` (Hart & Wang 2022). Linear projections are a safe way to view high-dimensional data because they do not warp the space, so they are more faithful representations of the structure. However, linear projections can be cluttered, and global patterns can obscure local structure. The simple activity of projecting data from p-D suffers from piling (Laa et al. 2022), where data concentrates in the center of projections. NLDR is designed to escape these issues, to exaggerate structure so that it can be observed. But as a result, NLDR can hallucinate wildly, to suggest patterns that are not actually present in the data.

Lee et al. (2022) provided interactive graphics to link between the NLDR view and tours, to help understand how the NLDR transformed the interpoint distances into a p-D layout. Here, we further develop this approach to directly use the tour to examine how the NLDR is warping the space. It follows what Wickham et al. (2015) describes as **model-in-the-data-space**. The fitted model should be overlaid on the data to examine the fit relative to the spread of the observations. While this is straightforward, and commonly done when data is 2-D, it is also possible in p-D, for many models, when a tour is used.

Wickham et al. (2015) provides several examples of models overlaid on the data in p-D. In hierarchical clustering, a representation of the dendrogram using points and lines can be constructed by augmenting the data with points marking the merging of clusters. Showing the movie of linear projections shows how the algorithm sequentially fitted the cluster model to the data. For linear discriminant analysis or model-based clustering, the model can be indicated by $(p-1)$-D ellipses. It is possible to see whether the elliptical shapes appropriately match the variance of the relevant clusters, and to compare and contrast different fits. For PCA, one can display the model (a k-D plane of the reduced dimension) using wireframes of transformed cubes. Using a wireframe is the approach we take here to represent the NLDR model in p-D.

3 Considering the NLDR layout as a model

At first glance, thinking of NLDR as a modeling technique might seem strange. It is a simplified representation or abstraction of a system, process, or phenomenon in the real world. The p-D observations are the realization of the phenomenon, and the k-D

NLDR layout is the simplified representation. Typically, $k = 2$ is used. From a statistical perspective, we can consider the distances between points in the 2-D layout to be variance that the model explains, and the (relative) difference with their distances in p-D as the error, or unexplained variance. We can also imagine that the positioning of points in 2-D represent the fitted values, which will have some prescribed position in p-D that can be compared with their observed values. This is the conceptual framework underlying the more formal versions of factor analysis (Jöreskog 1969) and MDS. (Note that, for this thinking, the full p-D data needs to be available, not just the interpoint distances.)

We define the NLDR as a function g: $\mathbb{R}^{n\times p} \rightarrow \mathbb{R}^{n\times 2}$, with hyper-parameters $\boldsymbol{\theta}$. These parameters, $\boldsymbol{\theta}$, depend on the choice of g, and can be considered part of model fitting in the traditional sense. Common choices for g include functions used in tSNE, UMAP, PHATE, TriMAP, PaCMAP, or MDS, although in theory any function that does this mapping is suitable.

With our goal being to make a representation of this 2-D layout that can be lifted into high-dimensional space, the layout needs to be augmented to include neighbor information. A simple approach would be to triangulate the points and add edges. A more stable approach is to first bin the data, reducing it from n to $m \leq n$ observations, and connect the bin centroids. We recommend using a hexagon grid because it better reflects the data distribution and has fewer artifacts than a rectangular grid (Carr et al. 1987). This process serves to reduce some noisiness in the resulting surface shown in p-D. The steps in this process are shown in Figure 2, and documented below.

To illustrate the method and how to use it to choose from a selection of layouts, we simulated

7-D data, which we call the "2NC7" data, that has two separated nonlinear clusters, one forming a 2-D curved shape (cluster 1), and the other a 3-D curved shape (cluster 2), each consisting of 1000 observations. The first four variables, X_1, X_2, X_3, X_4 hold this cluster structure, and the remaining three X_5, X_6, X_7 are low variance noise. Because, no error was simulated in the first four variables, we would consider $T = (X_1, X_2, X_3, X_4)$ to be **the geometric structure (true model)** that we hope to capture. In practice, all variables might have some small error, and structure to detect might be in any or all variables. The 2NC7 data is simple but complex enough illustrate the new methodology. We would consider the two separated clusters as *global structure*, and the nonlinear surfaces to be relatively *local structure*. Figure 2 uses an **reasonable** 2-D **NLDR layout** of 2NC7, produced by tSNE, which reveals the two clusters with moderate separation (global), and flattens the curvilinear forms while (hopefully) preserving the proximity of points (local). Note that, because cluster 2 is 3-D, it cannot be fully captured in a 2-D layout, and the flat rectangle is as good as can be expected.

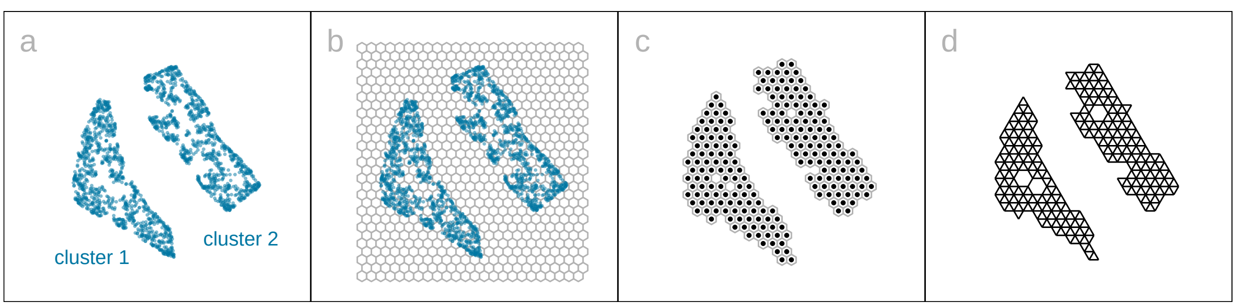

Figure 2: Key steps for constructing the model on the tSNE layout ($k = 2$): (a) data, (b) hexagon bins, (c) bin centroids, and (d) triangulated centroids. The layout used is a reasonable reprersntation of the 2NC7 data, which contains two curvilinear clusters that have been flattened into a triangular and rectangular shape.

4 Algorithm to represent the model in 2-D

4.1 Scale the layout data

Because the algorithm is based on distances between points, scaling the layout data to range between [0, 1] in the first coordinate, is recommended. By default, the aspect ratio produced by the NLDR $(r_1, r_2, ..., r_k)$ is preserved in the other coordinates. When $k = 2$, as in hexagon binning, the default range for the second coordinate is $[0, r_2/r_1]$.

4.2 Hexagon grid configuration

Although there are several implementations available (e.g., Carr et al. (2023)), the full set of data objects needed for this project. So we have re-implemented the method so that all items needed for creating the p-D representation are available.

The 2-D hexagon grid is uniquely defined by the hexagon ids, $h(= 1, \dots, b)$, and their 2-D centroids, $C_h^{(2)} = (c_{h1}, c_{h2})$. The algorithm inputs b_1, which determines b_2 using the NLDR ratio, producing $b = b_1 \times b_2$, the total number of bins in the grid.

A buffer is used to ensure that the grid covers the range of data values, and provide flexibility for where the the grid positining. It is well-known that where a grid starts can change binning results, even though most documentation focuses on binwidth! The buffer (q) is set as a proportion of the range. By default, $q = 0.1$, which is 10% of the range. The lower left position where the grid starts is set to be $(s_1 = -q, s_2 = -qr_2)$, specifying the centroid of the lower left hexagon, $C_1^{(2)} = (c_{11}, c_{12})$.

Considering the upper bound of the first NLDR component, the horizontal binwidth must satisfy $a_1 > (1 + 2q)/(b_1 - 1)$. Similarly, the vertical binwidth must satisfy

$$a_2 \geq \frac{r_2 + q(1 + r_2)}{(b_2 - 1)}. \tag{1}$$

For regular hexagons $a_2 = \sqrt{3}a_1/2$, Equation 1 can be re-written as

$$a_1 \geq \frac{2(r_2 + q(1 + r_2))}{\sqrt{3}(b_2 - 1)}.$$

This is a linear optimization problem, resulting in calculating b_2 as

$$b_2 = \left\lceil 1 + \frac{2(r_2 + q(1 + r_2)(b_1 - 1)}{\sqrt{3}(1 + 2q)} \right\rceil. \tag{2}$$

where $\lceil . \rceil$ indicate "smallest integer greater than or equal to".

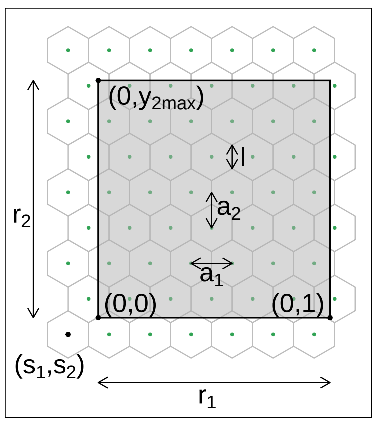

Figure 3: The components of the hexagon grid illustrating notation. The grey region indicates the extent of the data, which the grid must cover.

4.3 Binning the data

Observations are grouped into bins based on their nearest centroid. This produces a reduction in size of the data from n to m, where $m \leq b$ (total number of bins). This can be defined using the function $u : \mathbb{R}^{n\times 2} \rightarrow \mathbb{R}^{m\times 2}$, where

$$u(i) = \arg\min_{j=1,\dots,b} \sqrt{(y_{i1} - C^{(2)}_{j1})^2 + (y_{i2} - C^{(2)}_{j2})^2},$$

maps observation i into $H_h = \{i|u(i) = h\}$. The bin id, h, for each observation is needed to construct the model representation in high dimensions.

4.4 Indicating neighborhood

Delaunay triangulation (Lee & Schachter 1980, Gebhardt et al. 2024) is used to connect centroids to indicate neighboring observations in the NLDR layout (Figure 2 (d)). These are retained to preserve the neighborhood information when the model is lifted into p-D.

5 Rendering the model in p-D

The last step is to lift the 2-D model into p-D by computing p-D vectors that represent bin centroids. We use the p-D mean of the points in a given hexagon, H_h, denoted $C^{(p)}_h$, to map the centroid $C^{(2)}_h = (c_{h1}, c_{h2})$ to a point in p-D. Let the j^{th} component of the p-D mean be

$$C^{(p)}_{hj} = \frac{1}{n_h}\sum_{i=1}^{n_h} x_{hij}, \quad h = 1,\dots,b;\ j = 1,\dots,p;\ n_h > 0.$$

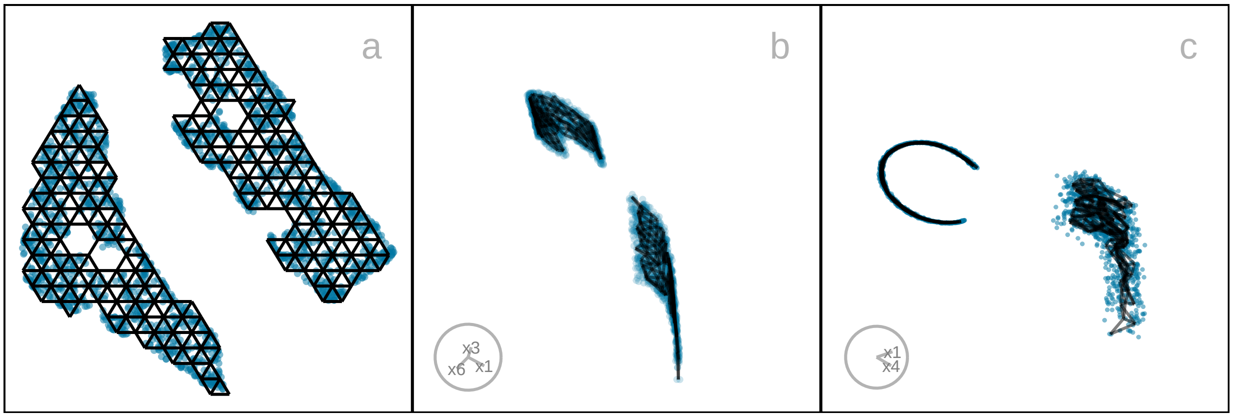

Figure 4: Lifting the 2-D fitted model into p-D. Two projections of the p-D fitted model overlaying the data are shown in b, c. The fit is reasonably tight with the data in one cluster (top one in b), but slightly less so in the other cluster, probably because it is 3-D. Notice also that, in the 2-D layout, the two clusters have internal gaps which creates a model with some holes. This lacy pattern happens regardless of the hyper-parameter choice, but this doesn't severely impact the p-D model representation.

6 Measuring the fit

All NLDR methods internally optimize an objective function to produce a layout for a given set of hyper-parameters. These objective functions are not always available in the model output and are generally not directly comparable across methods or hyper-parameter choices.

A range of established metrics are commonly used to assess the quality of NLDR embeddings, each emphasizing different aspects of structure preservation. For example, the RNX curve summarizes neighborhood agreement between p-D and k-D spaces across a range of scales, with its area under the curve (ARNX) reflecting a balance between local and global structure preservation (Lee et al. 2015). Random Triplet Accuracy (RTA) evaluates the consistency of distance orderings among triplets of points in 2-D and p-D spaces, providing a measure of geometric preservation (Wang et al. 2021). The Shepard diagram (Shepard 1962), together with its associated Spearman correlation (SC) (Spearman 1961), assesses the monotonic relationship between pairwise distances in the two spaces and is often interpreted as a measure of global structure preservation. The Global Score (GS) compares the embedding to a PCA baseline and evaluates how well the overall geometry is retained (Amid & Warmuth 2022).

These metrics provide useful but complementary perspectives, as they capture different and often competing objectives, such as preserving local neighborhoods versus global distances. As a result, it is common for these measures to rank embeddings differently, reflecting the multi-objective nature of NLDR rather than a deficiency of any single metric.

To facilitate comparison with our proposed measure, we use reversed forms of these metrics (rRTA, rSC, rGS, and rARNX), so that lower values consistently indicate better performance.

While the above measures evaluate properties of the embedding itself, our goal is different: to assess how well the embedding supports a representation of the data in the original $p\text{-}D$ space. This motivates a complementary approach based on model residuals.

We treat the embedding as defining a fitted representation in $p\text{-}D$, where each observation is associated with a bin centroid, $C_h^{(p)}$, which serves as its fitted value. The discrepancy between observed and fitted values can then be interpreted as a residual.

We define the hexbin error (HBE) as the root mean squared Euclidean distance between observations and their associated bin centroids:

$$HBE = \sqrt{\frac{1}{n}\sum_{h=1}^{m}\sum_{i=1}^{n_h}\sum_{j=1}^{p}(\boldsymbol{x}_{hij} - C_{hj}^{(p)})^2} \tag{3}$$

where n is the number of observations, m is the number of non-empty bins, n_h is the number of observations in h^{th} bin, p is the number of variables and $\boldsymbol{x}_{hij}$ is the j^{th} dimensional data of i^{th} observation in h^{th} hexagon. We can consider $e_{hi} = \sqrt{\sum_{j=1}^{p}(\boldsymbol{x}_{hij} - C_{hj}^{(p)})^2}$ to be the residual for each observation. Figure 5 visualizes these residuals as a density (a), as colors in the NLDR layout (b), and in a tour (c). Larger residuals are concentrated in one cluster, reflecting the intentional design that this cluster has intrinsic higher-dimensional structure (approximately 3-D) and therefore cannot be fully represented in a 2-D embedding. Importantly, HBE is not intended to replace existing embedding-quality metrics. Instead, it provides a complementary diagnostic perspective, highlighting where the embedding fails to represent the original data structure in p-D. Differences between HBE and other metrics can therefore be interpreted as informative signals about trade-offs between local and global structure preservation.

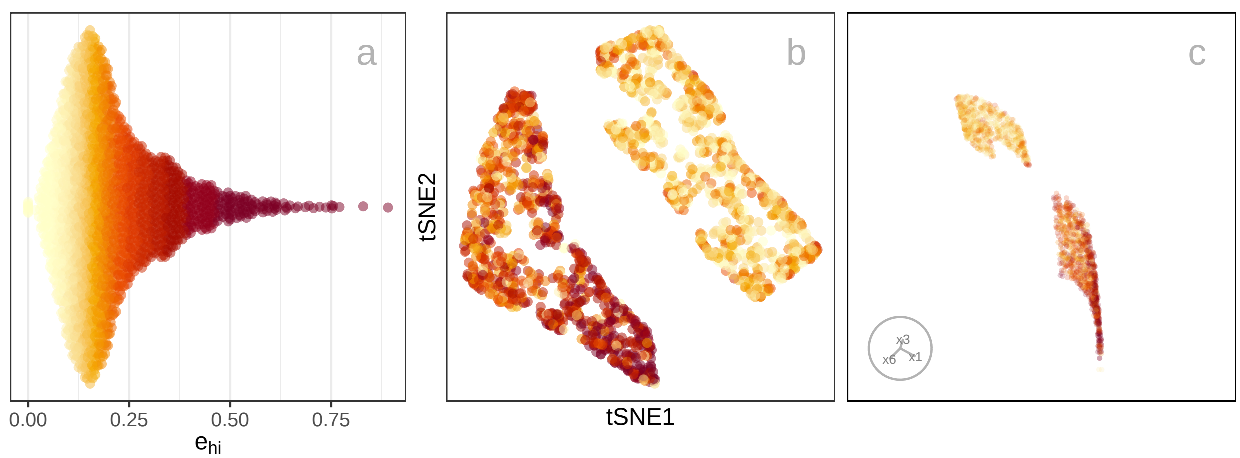

Figure 5: Examining the distribution of residuals in a jittered dotplot (a), 2-D NLDR layout (b), and a tour of 4-D data space (c). Color indicates residual (e_{hi}), dark color indicating high value. Most large residuals are distributed in one cluster (bottom one in c), and most small residuals are distributed in the other cluster.

7 Prediction into 2-D

Starting with a good layout produced by UMAP- maybe this should be the configuration, or something similar used throughout the first part of the paper.

NLDR methods focus on a visualization solution for exploration. Only UMAP provides a way to predict the layout positions of new observations (Konopka 2023). Other approaches not illustrated in this work, such as neural network autoencoders (Hinton & Salakhutdinov 2006), and parametric tSNE (van der Maaten 2009), provide some mechanisms for embedding new data. A benefit of our approach, is that with the model representation, we can predict layout positions for any method.

The procedure isfor the new observation: (i) identify the closest bin centroid in p-D, $C_h^{(p)}$, and (ii) assign the corresponding centroid in 2-D, $C_h^{(2)}$, as the predicted embedding. This is not entirely different from how UMAP provides predictions: find the nearest neighbors in p-D, and interpolate their positions in 2-D to provide the predicted embedding.

Figure 6 shows the results for UMAP (a) and our predictions(b) made using `quollr`. The `2NC7` data is divided into training and test sets, using a 70% split. The training data is used to make the 2-D layout. The test set is held out for prediction. In both cases it can be seen that the test set matches the training set in the layout quite well. The main differences is that our predictions are to the bin centroid, so it looks gridded. Adding some jitter to the predicted values would make them look more similar to the UMAP predictions. Just because we can, doesn't mean we should - for this data, the method provides demonstrably reasonable results, and better than having no method at all. **Add tSNE, PHATE, TriMAP and PACMAP training and quollr predictions to this plot, in a 3x2**

layout

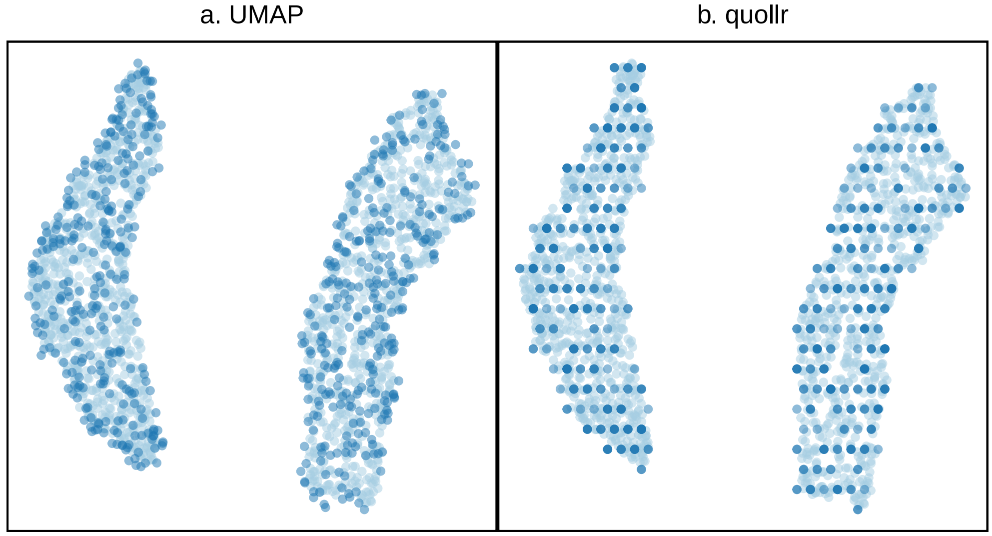

Figure 6: Comparison of predictions made my UMAP and by our method, relative to the training data. Dark points are the predicted test set, and lighter points are the training sample used to create the UMAP layout.

8 Interactive graphics

Matching points in the 2-D layout with their positions in p-D is useful when tuning the fit. This can be used to examine the fitted model in some subspaces in p-D, in particular in association with residual plots.

The interactive 2-D layout (Sievert 2020) and the langevitour (Harrison 2023) view with the fitted model overlaid can be linked using a browsable HTML widget (Cheng & Sievert (2025), Cheng et al. (2024)). A rectangular "brush" is used to select points in one plot, which will highlight the corresponding points in the other plot(s). Because the langevitour is dynamic, brush events that become active will pause the animation, so that a user can interrogate the current view. This approach will be illustrated on the examples, to show how it can help to understand how the NLDR has organized the observations and learn where it does not do well.

9 Choosing from a selection of layouts

Figure 7 illustrates the approach to compare the fits for different representations and assess the strength of any fit. What does it mean to be a best fit for this problem? Analysts use an NLDR layout to display the structure present in high-dimensional data in a convenient 2-D display. It is a competitor to linear dimension reduction that can better represent nonlinear associations, such as clusters. However, these methods can hallucinate, suggesting patterns that don't exist, and grossly exaggerate other patterns. Having a layout that best fits the high-dimensional structure is desirable, but more important is to identify bad representations so they can be avoided. The goal is to help users decide on the most useful and appropriate low-dimensional representation of the high-dimensional data.

A particular pattern that we commonly see is that analysts tend to pick layouts with clusters that have big separations between them. When you examine their data in a tour, it is almost always that we see there are no big separations, and actually often the suggested clusters are not even present. While we don't expect that analysts include animated gifs of tours in their papers, we should expect that any 2-D representation adequately indicates the clustering that is present, and honestly show lack of separation or lack of clustering when it doesn't exist. It is important for analysts to have tools to select the accurate representation, not the pretty but wrong representation.

Figure 7 compares the metrics rARNX, rRTA, rSC, rGS, along with HBE computed on $a_1 = 0.05$ for the six layouts shown in Figure 7. This is a parallel coordinate plot where

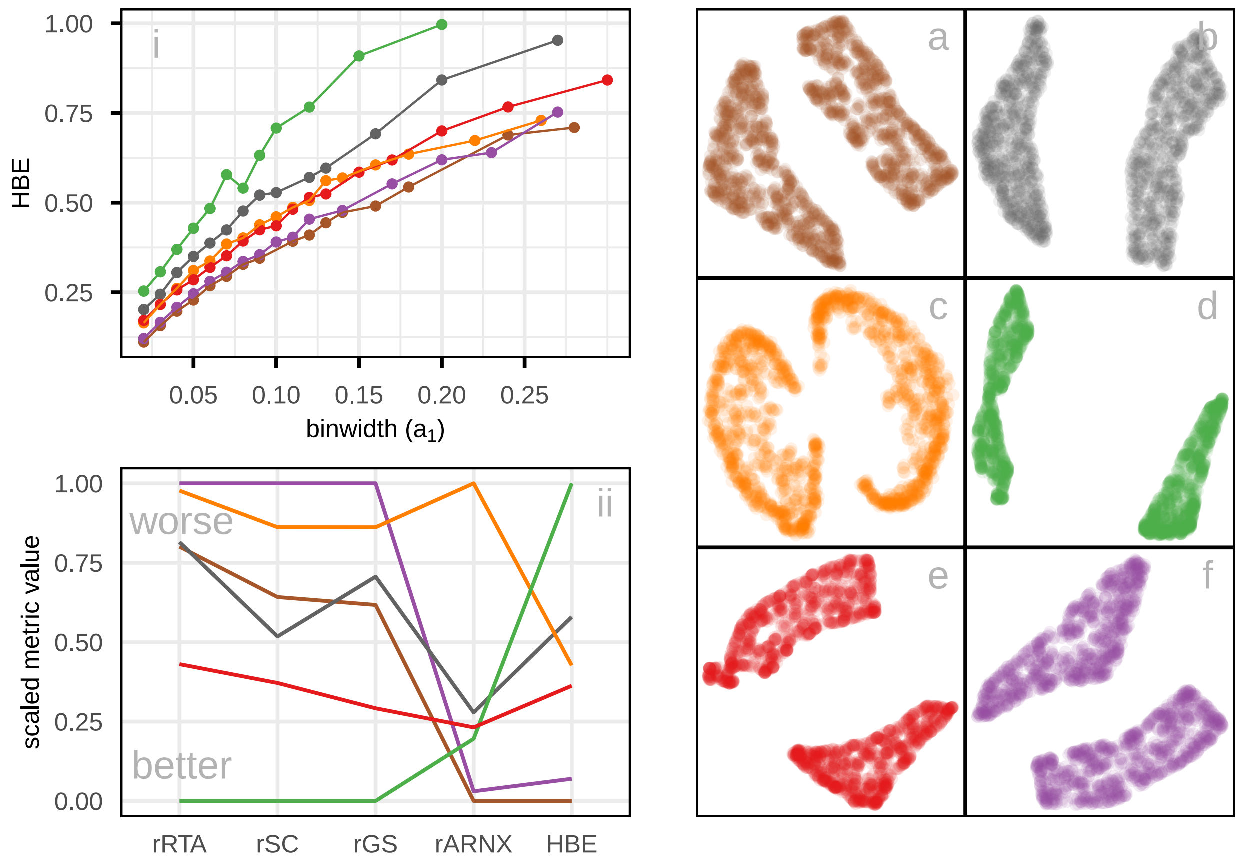

Figure 7: Assessing which of the 6 NLDR layouts (a-f) on the 2NC7 data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color represents the NLDR layout. Layout d is universally poor. Layouts b and e that show two close clusters are universally suboptimal. Layout f with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. Layout e has a small separation with oddly shaped clusters. Layout a is the best choice. Plot (ii), which compares HBE values with respect to average bin count, helps account for differences in cluster density; here, the variation among layouts is reduced, showing that some differences observed in (i) arise from density rather than true structure. Comparison of scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE using $a_1 = 0.05$) for the six NLDR layouts computed on the 2NC7 data using a parallel coordinate plot (iii). Color of the line indicates NLDR layout.

the y-axis shows a normalized score to ensure the metrics are on the same scale. Each line corresponds to one layout.

There is some agreement between the metrics. All, except rARNX and HBE, agree that layout d is best. rARNX and HBE agree that layout f is best or very close to best. Layout a is best according to HBE and rARNX but considered to be much less optimal by rRTA, rSC and rGS. Layout c is considered poor by rRTA, rSC, rGS, and rARNX. This illustrates how difficult it is to use the numerical metrics alone to decide on the best layout.

The problem with rSC is that correlation is not a good measure in the presence of clusters - the further the clusters are apart in the layout, produces a Shepard plot with two clusters of distances, which will produce a high correlation value. Similar reasoning would explain why rRTA and rGS behave similarly: they put too much emphasis on the global structure. Thus, for the 2NC7 data, further apart clusters score better, overly emphasizing that there are two clusters, even though this separation is not accurately reflecting the difference in p-D.

When the metrics disagree is causes confusion for the analyst, and thus provides a temptation to choose the nicest looking layout (very separated clusters), even though it may be a hallucination. Because HBE is accompanied with a representation of the layout in p-D to compare with the observed data, it can help to add more clarity in making decisions. Figure 8 shows the fitted models for layouts a (rated high by HBE and rARNX) and c (rated poorly). These are 2-D projections from the tour, with black indicating the fitted model overlaid on the blue points of the data. The reason for the poor fit is that the PHATE layout (c) twists extremely along the 2- and 3-D manifolds where the data lies. We have learned that all the NLDR methods tend to have twists in the fit in p-D, but this is extreme.

This is likely why layout c has poor metrics relative to the other layouts, and it suggests that it does not adequately capture the local structure in the 2NC7 data.

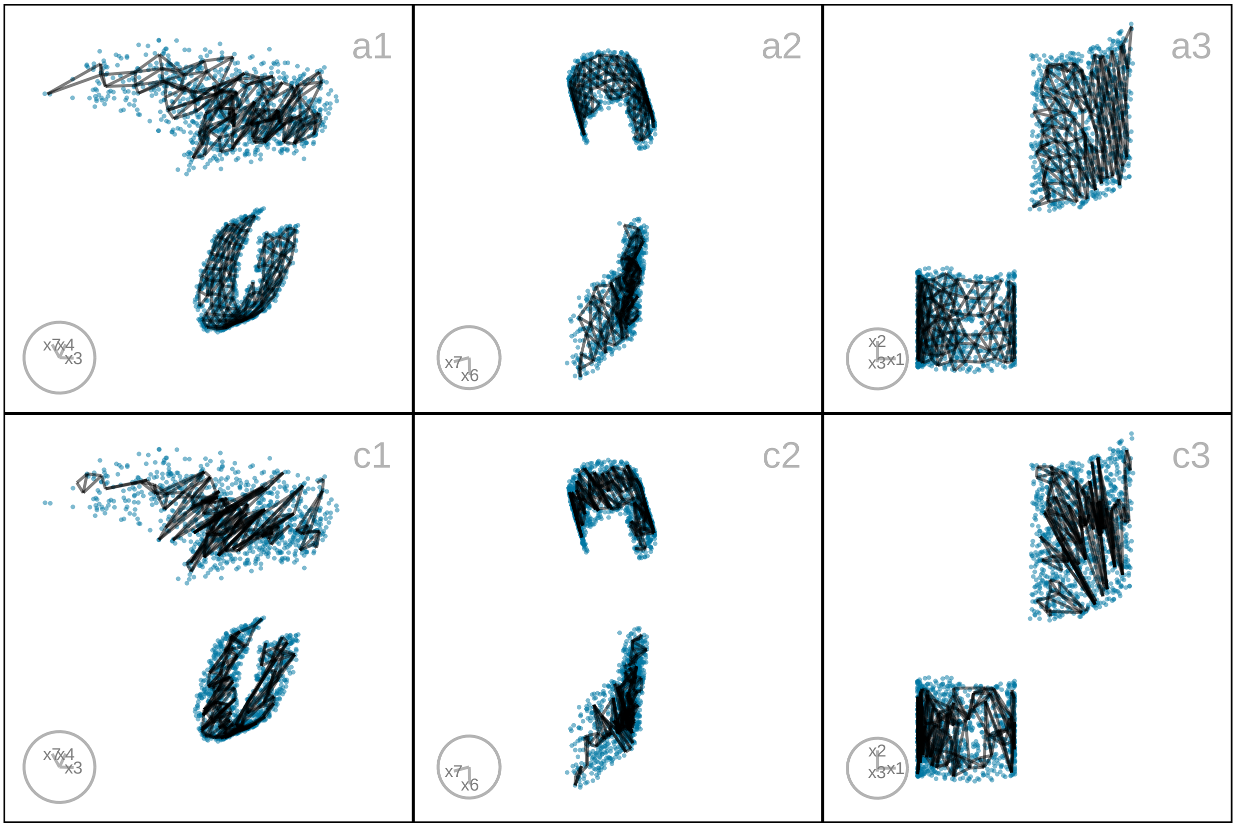

Figure 8: Three 2-D projections from a tour showing the fitted models (black lines) for layouts a (top row) and c (bottom row) of the 2NC7 data (blue points). Layout c, which was poorly rated by rARNX and HBE, covers less of the width of the data than layout a. The triangular gridding is less visible in c than layout a, and actually corresponds to extreme twisting.

10 Applications

To illustrate the approach, we use two examples: PBMC3k data (single cell gene expression), where an NLDR layout is used to represent cluster structure present in the p-D data, and MNIST hand-written digits, where NLDR is used to represent a low-dimensional nonlinear manifold in p-D.

10.1 PBMC3k

This is a benchmark single-cell RNA-Seq data set collected on Human Peripheral Blood Mononuclear Cells (PBMC3k) as used in 10x Genomics (2016). Single-cell data measures the gene expression of individual cells in a sample of tissue (see, for example, Haque et al. (2017)). This type of data is used to obtain an understanding of cellular level behavior and heterogeneity in their activity. Clustering of single-cell data is used to identify groups of cells with similar expression profiles. NLDR is often used to summarize the cluster structure. Usually, NLDR does not use the cluster labels to compute the layout, but uses color to represent the cluster labels when it is plotted.

In this data, there are 2622 single cells and 1000 gene expressions (variables). Following the same pre-processing as Chen et al. (2024), different NLDR techniques were performed on the first nine principal components. Figure 1 shows this data using a variety of methods and different hyper-parameters. You can see that the result is wildly different depending on the choices. Layout a is a reproduction of the layout that was published in Chen et al. (2024). This layout suggests that the data has three very well-separated clusters, each with an odd shape. The question is whether this accurately represents the cluster structure in the data, or whether they should have chosen b or c or d or e or f or g or h. This is what our new method can help with – to decide which is the more accurate 2-D representation of the cluster structure in the p-D data.

Figure 9 shows HBE across a range of binwidths (a_1) for each of the layouts in Figure 1. The layouts were generated using tSNE and UMAP with various hyper-parameter settings, while PHATE, PaCMAP, and TriMAP were applied using their default settings. Lines are color coded to match the color of the layouts shown on the right. Lower HBE indicates a better fit. Using a range of binwidths shows how the model changes, with possibly the best model being one that is universally low HBE across all binwidths. It can be seen that layout f is sub-optimal with universally higher HBE. Layout a, the published one, is better, but it is not as good as layouts b, d, or e. With some imagination, the layout d perhaps shows three barely distinguishable clusters. Layout e shows three, possibly four, clusters that are more separated. The choice reduces from eight to these two. Layout d has slightly better HBE when the a_1 is small, but layout e beats it at larger values. Thus, we could argue that layout e is the most accurate representation of the cluster structure of these eight.

To further assess the choices, we need to look at the model in the data space, by using a tour to show the wireframe model overlaid on the data in the 9-D space (Figure 10). Here we compare the published layout (a) versus what we argue is the best layout (e). The top row (a1, a2, a3) corresponds to the published layout, and the bottom row (e1, e2, e3) corresponds to the optimal choice according to our procedure. The middle and right plots show two projections. The primary difference between the two models is that the model of layout e does not fill out to the extent of the data but concentrates in the center of each point cloud. Both suggest that three clusters are a reasonable interpretation of the structure, but the layout e more accurately reflects the separation between them, which is small.

10.2 MNIST hand-written digits

The digit "1" of the MNIST dataset (LeCun et al. 1998) consists of 7877 grayscale images of handwritten "1"s. Each image is 28×28 pixels, which corresponds to 784 variables. The

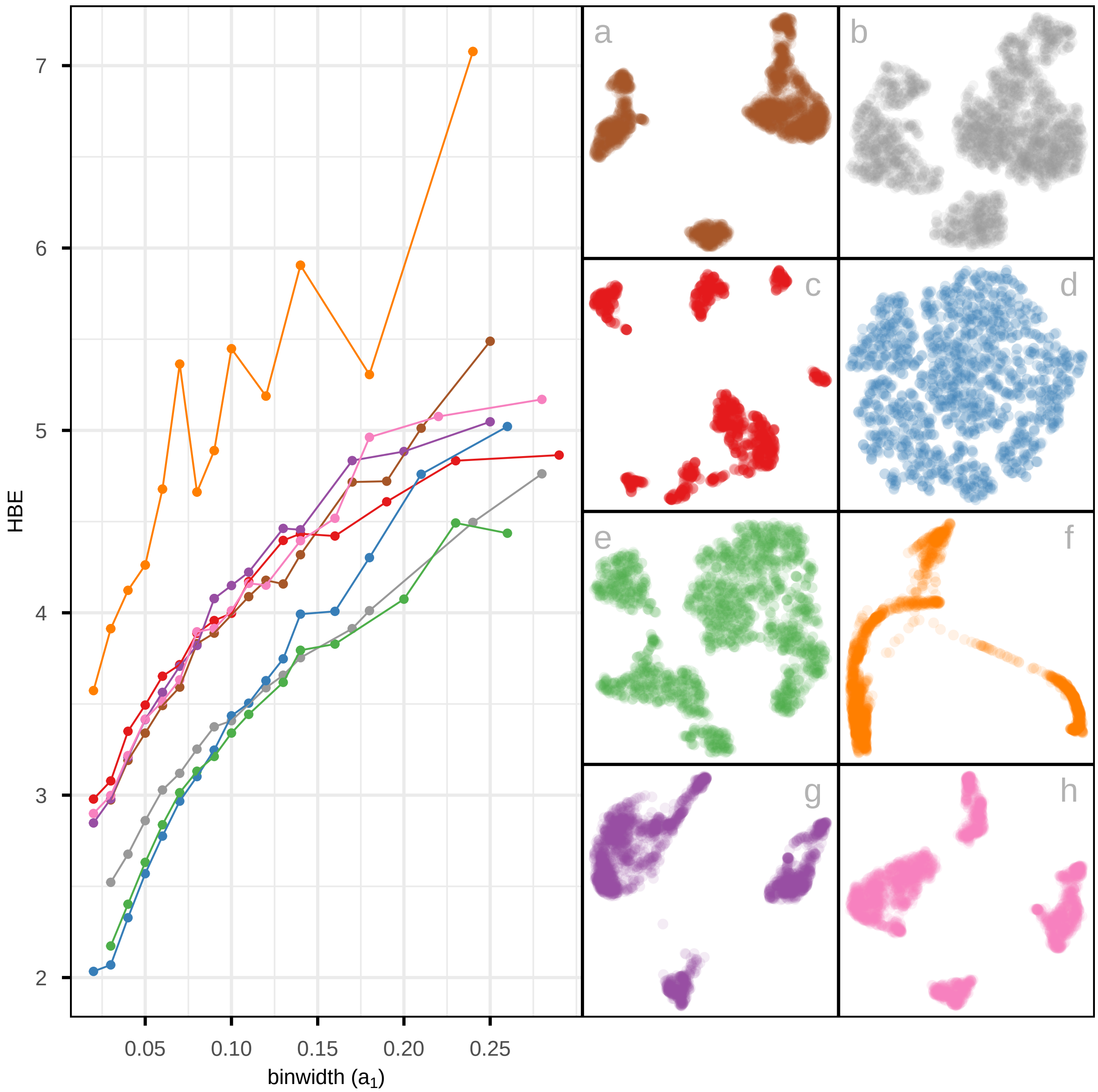

Figure 9: Assessing which of the 8 NLDR layouts on the PBMC3k data (shown in Figure 1) is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color used for the lines and points in the left plot and in the scatterplots represents NLDR layout (a-h). Layout f is universally poor. Layouts a, c, g, and h that show large separations between clusters are universally suboptimal. Layout d with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. The choice of the best is between layouts b and e, which have small separations between oddly shaped clusters. Layout e is chosen as the best. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

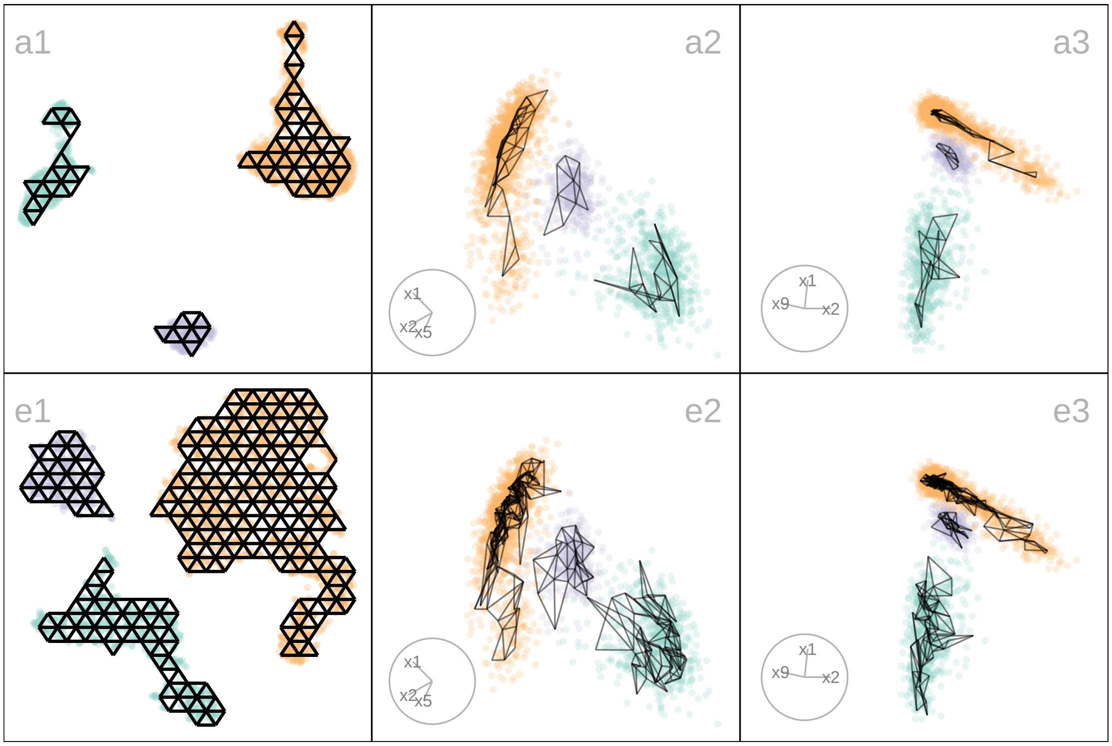

Figure 10: Compare the published 2-D layout (a) made with UMAP and the 2-D layout selected by HBE plot (e) made by tSNE. The two plots on the right show projections from a tour, with the models overlaid. The published layout a suggested three very separated clusters, but this is not present in the data. While there may be three clusters, they are not well-separated. The difference in model fit also indicates this: the published layout a does not spread out fully into the point cloud like the model generated from layout e. This supports the choice that layout e is the better representation of the data, because it does not exaggerate separation between clusters.

first 10 principal components, explaining 83% of the total variation, are used. This data essentially lies on a nonlinear manifold in high dimensions, defined by the shapes that "1"s make when sketched. We expect that the best layout captures this type of structure and does not exhibit distinct clusters.

Figure 11 compares the fit of six layouts computed using UMAP (b), PHATE (c), TriMAP (d), PaCMAP (e) with default hyper-parameter setting and two tSNE runs, one with default hyper-parameter setting (a) and the other changing perplexity to 89 (f). The layouts are reasonably similar in that they all have the observations in a single blob. Some (b, c) have

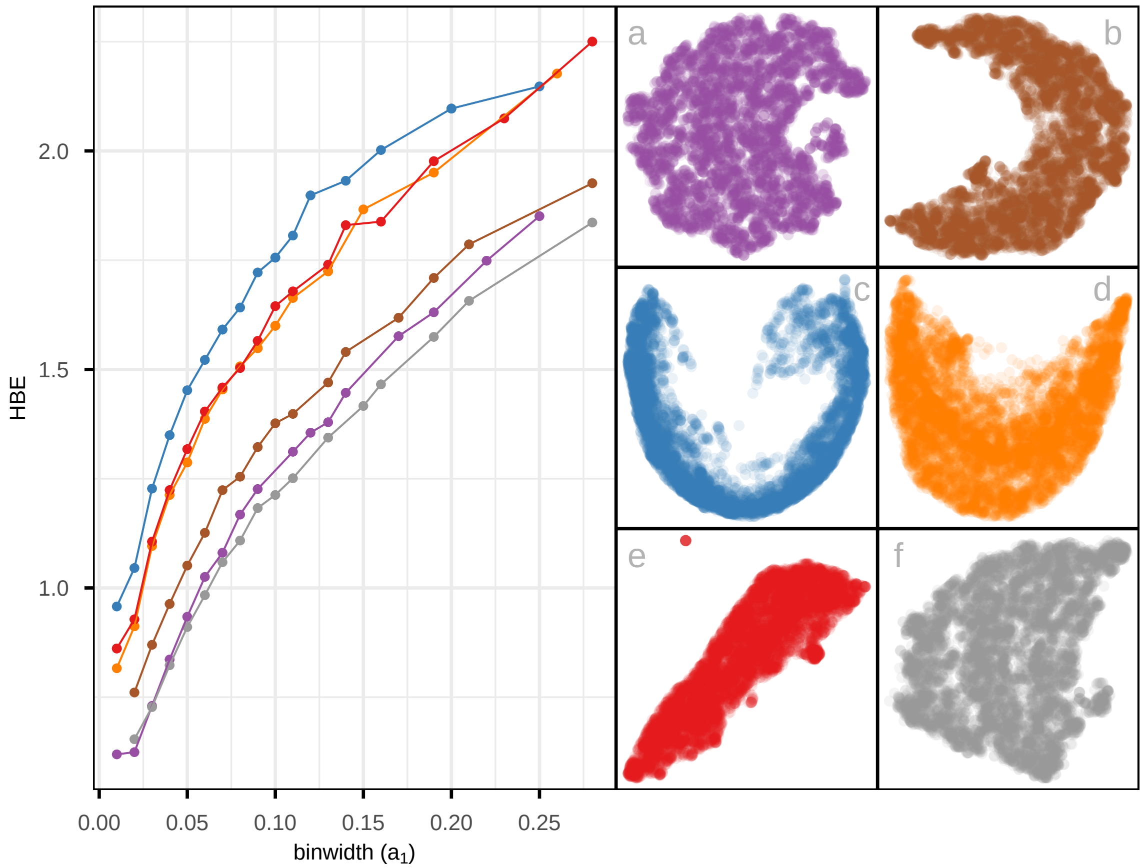

Figure 11: Assessing which of the 6 NLDR layouts of the MNIST digit 1 data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color is used for the lines and points in the left plot to match the scatterplots of the NLDR layouts (a-f). Layout c is universally poor. Layouts a and f that show a big cluster and a small circular cluster are universally optimal. Layout a performs well at tiny binwidth (where most points are in their own bin) and not as well as f with larger binwidth, thus layout f is the best choice. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

a more curved shape than others. Layout e is the most different, having a linear shape and a single very large outlier. Both a and f have a small clump of points, perhaps slightly disconnected from the other points, in the lower to middle right.

The layout plots are colored to match the lines in the HBE vs binwidth (a_1) plot. Layouts a, b, and f fit the data better than c, d, e, and layout f appears to be the best fit. Figure 12 shows this model in the data space in two projections from a tour. The data is curved in the 10-D space, and the fitted model captures this curve. The small clump of points in the 2-D layout is highlighted in both displays. These are almost all inside the curve of the bulk of points and are sparsely located. The fact that they are packed together in the 2-D layout is likely due to the handling of density differences by the NLDR.

An interesting aside is that the rather strange layout e, which has what looks like a single point far from the remaining observations, is actually similar to this one. That point is actually a clump of points corresponding to some of the diffuse points interior to the curve of the bulk of points. This is easy to see using the linked brushing tool.

The next step is to investigate the 2-D layout to understand what information is learned from this representation. Figure 13 summarizes this investigation. Plot a shows the layout with points colored by their residual value - darker color indicates larger residual and poor fit. The plots b, c, d, e show samples of hand-written digits taken from inside the colored boxes. Going from top to bottom around the curve shape, we can see that the “1”s are drawn from right slant to a left slant. The “1”s in d (black box) tend to have the extra up stroke, but are quite varied in appearance. The “1”s shown in the plots labelled e correspond to points with big residuals. They can be seen to be more strangely drawn than the others. Overall, this 2-D layout shows a useful way to summarize the variation in ways “1”s are drawn.

11 Discussion

We have developed an approach to help assess and compare NLDR layouts, generated by different methods and hyper-parameter choice(s). It depends on conceptualizing the 2-D

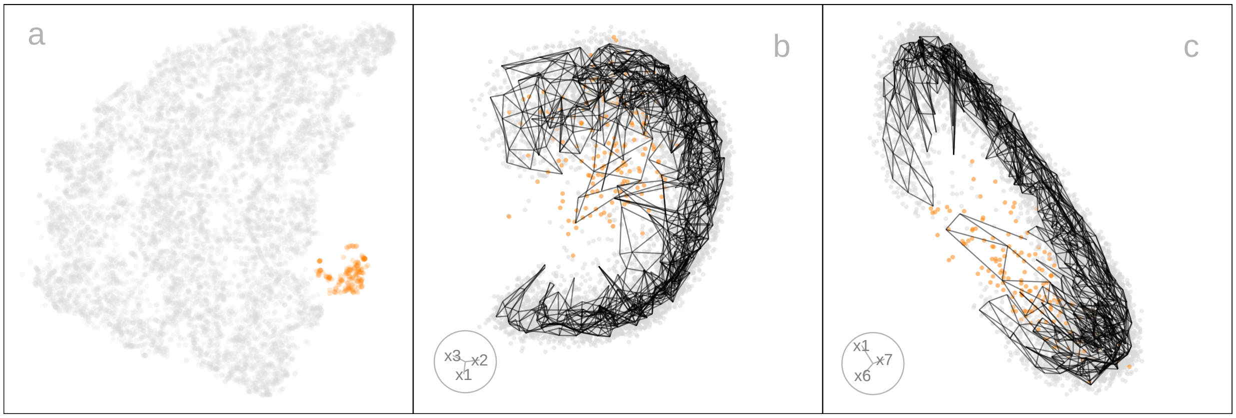

Figure 12: Summary from exploring tSNE layout of the MNIST digit 1 data (a in Figure 12) using linked brushing. There is a big nonlinear cluster (grey) and a small cluster (orange) located very close to one corner of the big cluster in 2-D (a). The MNIST digit 1 data has a nonlinear structure in 10-D. Two 2-D projections from a tour on 10-D reveal that the small orange cluster is actually a diffuse set of points wrapped within the grey cluster, which is C-shaped in the high dimensions.

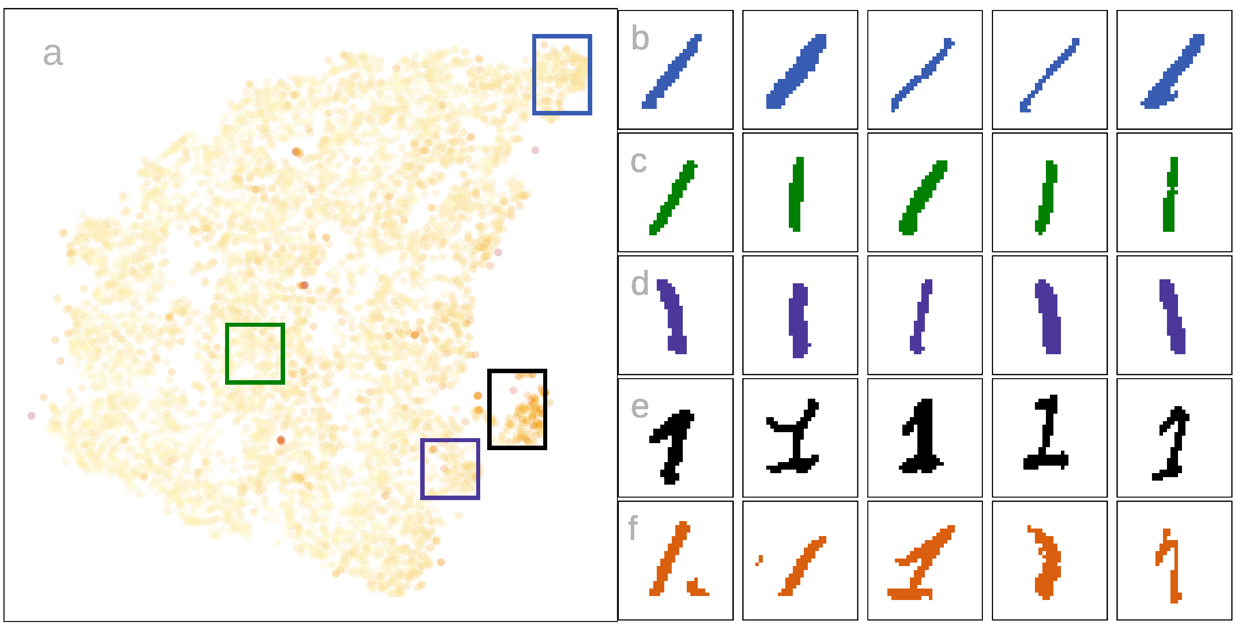

Figure 13: Summary of the layout structure, and large errors, relative to the MNIST digit 1 shape: (a) layout colored by residual value, and at right (b-e) are images of samples of observations taken at locations around the layout, showing similarity in how the 1's were drawn. Set (f) are images corresponding to large residuals in the big cluster (darker orange in plot a). Along the big cluster, the angle of digit 1 changes (b-d). The small cluster has larger residuals, and the images show that these tend to be European style with a flag at the top and a base at the bottom. The set in (f) shows various poorly written digits.

layout as a model, allowing for the creation of a wireframe representation of the model that can be lifted into p-D. The fit is assessed by viewing the model in the data space, computing residuals, and HBE. Different layouts can be compared using the HBE, providing a quantitative metric to decide on the most suitable NLDR layout to represent the p-D data. Global and local preservation of structure is assessed by examining the HBE across a range of binwidths. It also provides a way to predict the values of new p-D observations in the 2-D, which could be useful for implementing uncertainty checks, such as using training and testing samples.

The new methodology is accompanied by an R package called `quollr`, so that it is readily usable and broadly accessible. The package has methods to fit the model, compute diagnostics, and also visualize the results, with interactivity. We have primarily used the `langevitour` software (Harrison 2023) to view the model in the data space, but other tour software such as `tourr` (Wickham et al. 2011) and `detourr` (Hart & Wang 2022) could also be used.

Two examples illustrating usage are provided: the PBMC3k data, where the NLDR is summarizing clustering in p-D, and hand-written digits illustrating how NLDR represents an intrinsically low-dimensional nonlinear manifold. We examined a typical published usage of UMAP with the PBMC3k dataset (Chen et al. 2024). As is typical of UMAP layout with default settings, the separation between clusters is grossly exaggerated. The layout even suggests separation where there is none. Our approach provides a way to choose a reasonable layout and avoids the use of misleading layouts in the future. In the hand-written digits (LeCun et al. 1998), we illustrate how our model fit statistics show that a flat disc layout is superior to the curved-shaped layouts, and how to identify oddly written “1”s using the residuals of the fitted model.

This work can be applied with existing metrics for evaluating NLDR layout, such as ARNX, RTA, SC, and RGS. It provides an additional evaluation metric, and importantly allows any layout to be viewed in the p-D data space. This latter aspect can help to disentangle conflicting suggestions by the different metrics.

Additional exploration of distance measures to summarize the fit could be a valuable direction for future work. We have used Euclidean distance, but other measures, such as geodesic distances (Tenenbaum et al. 2000), may better capture curved or nonlinear relationships in the data and are worth exploring.

This work has also revealed some interesting behaviors of NLDR methods, including twisting, flattened "pancake" clusters in p-D, and severe effects of density differences. These are described in more detail in the supplementary materials.

Researchers usually use 2-D layouts, but if a k-D ($k > 2$) layout is provided, the approach developed here could be extended. Potential approaches include 3-D binning, k-means clustering, or even special implementations of convex hulls.

12 Supplementary Materials

All the materials to reproduce the paper can be found at https://github.com/JayaniLakshika/paper-nldr-vis-algorithm.

The supplementary materials provide additional details on the methods and hyper-parameters used to generate layouts, video links of animated p-D tours, notation summaries, and the R and Python scripts used in the study. They also describe the generation of the 2NC7 data, computation of hexagon grid configurations, and data binning procedures.

Further sections highlight interesting NLDR behaviors observed in the data space and compare HBE with existing evaluation metrics for the PBMC3k and MNIST datasets.

The R package `quollr`, available on CRAN and at https://jayanilakshika.github.io/quollr/, provides software accompanying this paper to fit the wireframe model representation, compute diagnostics, visualize the model in the data with `langevitour`, and link multiple plots interactively.

13 Acknowledgments

These `R` packages were used for the work: `tidyverse` (Wickham et al. 2019), `Rtsne` (Krijthe 2015), `umap` (Konopka 2023), `patchwork` (Pedersen 2024), `colorspace` (Zeileis et al. 2020), `langevitour` (Harrison 2023), `conflicted` (Wickham 2023), `reticulate` (Ushey et al. 2024), `kableExtra` (Zhu 2024). These `python` packages were used for the work: `trimap` (Amid & Warmuth 2022) and `pacmap` (Wang et al. 2021). The article was created with `R` packages `quarto` (Allaire & Dervieux 2024).

References

10x Genomics (2016), '3k PBMCs from a Healthy Donor, Universal 3' Gene Expression dataset analysed using Cell Ranger 1.1.0'. Accessed: 2025-06-09. https://www.10xgenomics.com/datasets/3-k-pbm-cs-from-a-healthy-donor-1-standard-1-1-0.

Allaire, J. & Dervieux, C. (2024), *quarto: R Interface to Quarto Markdown Publishing System*. R package version 1.4.4. https://CRAN.R-project.org/package=quarto.

Amid, E. & Warmuth, M. K. (2022), 'TriMap: Large-scale Dimensionality Reduction Using Triplets', *arXiv preprint arXiv:1910.00204* .

Asimov, D. (1985), 'The Grand Tour: A Tool for Viewing Multidimensional Data', *SIAM Journal of Scientific and Statistical Computing* **6**(1), 128–143.

Borg, I. & Groenen, P. J. F. (2005), *Modern Multidimensional Scaling*, 2nd edn, Springer, New York, NY. https://doi.org/10.1007/0-387-28981-X.

Carr, D. B., Littlefield, R. J., Nicholson, W. L. & Littlefield, J. S. (1987), 'Scatterplot Matrix Techniques for Large N', *Journal of the American Statistical Association* **82**(398), 424–436. http://www.jstor.org/stable/2289444.

Carr, D., Lewin-Koh, N., Maechler, M. & Sarkar, D. (2023), *hexbin: Hexagonal Binning Routines*. R package version 1.28.3. https://CRAN.R-project.org/package=hexbin.

Chari, T. & Pachter, L. (2023), 'The specious art of single-cell genomics', *PLoS Computational Biology* **19**, e1011288. https://doi.org/10.1371/journal.pcbi.1011288

Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), 'Directly Selecting Cell-type Marker Genes for Single-cell Clustering Analyses', *Cell Reports Methods* **4**(7), 100810. https://www.sciencedirect.com/science/article/pii/S2667237524001735.

Cheng, J. & Sievert, C. (2025), *crosstalk: Inter-Widget Interactivity for HTML Widgets*. R package version 1.2.2. https://CRAN.R-project.org/package=crosstalk

Cheng, J., Sievert, C., Schloerke, B., Chang, W., Xie, Y. & Allen, J. (2024), *htmltools: Tools for HTML*. R package version 0.5.8.1. https://CRAN.R-project.org/package=htmltools

Coifman, R. R., Lafon, S., Lee, A. B., Maggioni, M., Nadler, B., Warner, F. & Zucker, S. W.

(2005), 'Geometric Diffusions as a Tool for Harmonic Analysis and Structure Definition of Data: Diffusion Maps', *Proceedings of the National Academy of Sciences of the United States of America* **102**(21), 7426–7431. https://www.pnas.org/doi/abs/10.1073/pnas.0500334102.

Gebhardt, A., Bivand, R. & Sinclair, D. (2024), *interp: Interpolation Methods*. R package version 1.1-6. https://CRAN.R-project.org/package=interp.

Haque, A., Engel, J., Teichmann, S. A. & Lönnberg, T. (2017), 'A Practical Guide to Single-cell RNA-sequencing for Biomedical Research and Clinical Applications', *Genome Medicine* **9**(1), 75. https://doi.org/10.1186/s13073-017-0467-4.

Harrison, P. (2023), 'langevitour: Smooth Interactive Touring of High Dimensions, Demonstrated With scRNA-Seq Data', *The R Journal* **15**, 206–219. https://doi.org/10.32614/RJ-2023-046.

Hart, C. & Wang, E. (2022), *detourr: Portable and Performant Tour Animations*. R package version 0.1.0. https://casperhart.github.io/detourr/.

Hinton, G. E. & Salakhutdinov, R. R. (2006), 'Reducing the dimensionality of data with neural networks', *Science* **313**(5786), 504–507.

Irizarry, R. (2024), 'Biologists, stop putting umap plots in your papers', https://simplystatistics.org/posts/2024-12-23-biologists-stop-including-umap-plots-in-your-papers/. Blog post on Simply Statistics.

Johnstone, I. M. & Titterington, D. M. (2009), 'Statistical Challenges of High-Dimensional Data', *Philosophical Transactions of the Royal Society A: Mathematical, Physical and*

Engineering Sciences **367**(1906), 4237–4253. https://royalsocietypublishing.org/doi/abs/10.1098/rsta.2009.0159.

Jolliffe, I. (2011), *Principal Component Analysis*, Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 1094–1096. https://doi.org/10.1007/978-3-642-04898-2_455.

Jöreskog, K. G. (1969), 'A General Approach to Confirmatory Maximum Likelihood Factor Analysis', *Psychometrika* **34**(2), 183–202. https://doi.org/10.1007/BF02289343.

Koch, I. (2013), *Analysis of Multivariate and High-Dimensional Data*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press.

Konopka, T. (2023), *umap: Uniform Manifold Approximation and Projection*. R package version 0.2.10.0. https://CRAN.R-project.org/package=umap.

Krijthe, J. H. (2015), *Rtsne: T-Distributed Stochastic Neighbor Embedding Using Barnes-Hut Implementation*. R package version 0.16. https://github.com/jkrijthe/Rtsne.

Kruskal, J. B. (1964), 'Nonmetric Multidimensional Scaling: A Numerical Method', *Psychometrika* **29**(2), 115–129. https://doi.org/10.1007/BF02289694.

Laa, U., Cook, D. & Lee, S. (2022), 'Burning Sage: Reversing the Curse of Dimensionality in the Visualization of High-Dimensional Data', *Journal of Computational and Graphical Statistics* **31**(1), 40–49. https://doi.org/10.1080/10618600.2021.1963264.

LeCun, Y., Cortes, C. & Burges, C. J. C. (1998), 'The MNIST Database of Handwritten Digits'. Accessed: 2025-06-09. http://yann.lecun.com/exdb/mnist/.

Lee, D. T. & Schachter, B. J. (1980), 'Two Algorithms For Constructing a Delaunay

Triangulation', *International Journal of Computer & Information Sciences* **9**(3), 219–242. https://doi.org/10.1007/BF00977785.

Lee, J. A., Peluffo-Ordóñez, D. H. & Verleysen, M. (2015), 'Multi-scale similarities in stochastic neighbour embedding: Reducing dimensionality while preserving both local and global structure', *Neurocomputing* **169**, 246–261. https://www.sciencedirect.com/science/article/pii/S0925231215003641

Lee, S., Cook, D., da Silva, N., Laa, U., Wang, E., Spyrison, N. & Zhang, H. S. (2021), 'A Review of the State-of-the-Art on Tours for Dynamic Visualization of High-Dimensional Data', *arXiv preprint arXiv:2104.08016* .

Lee, S., Laa, U. & Cook, D. (2022), 'Casting multiple shadows: interactive data visualisation with tours and embeddings', *Journal of Data Science, Statistics, and Visualisation* **2**(3). https://jdssv.org/index.php/jdssv/article/view/21

Maaten, L. V. D. & Hinton, G. E. (2008), 'Visualizing Data Using t-SNE', *Journal of Machine Learning Research* **9**(11), 2579–2605. https://api.semanticscholar.org/CorpusID:5855042.

McInnes, L., Healy, J., Saul, N. & Großberger, L. (2018), 'UMAP: Uniform Manifold Approximation and Projection', *Journal of Open Source Software* **3**(29), 861. https://doi.org/10.21105/joss.00861.

Moon, K. R., van Dijk, D., Wang, Z., Gigante, S. A., Burkhardt, D. B., Chen, W. S., Yim, K., van den Elzen, A., Hirn, M. J., Coifman, R. R., Ivanova, N. B., Wolf, G. & Krishnaswamy, S. (2019), 'Visualizing Structure and Transitions in High-Dimensional Biological Data', *Nature Biotechnology* **37**, 1482–1492. https://doi.org/10.1038/s41587-019-0336-3.

Pedersen, T. L. (2024), *patchwork: The Composer of Plots*. R package version 1.2.0. https://CRAN.R-project.org/package=patchwork.

Saeed, N., Nam, H., Haq, M. I. U. & Muhammad Saqib, D. B. (2018), 'A Survey on Multidimensional Scaling', *ACM Comput. Surv.* **51**(3). https://doi.org/10.1145/3178155.

Shepard, R. N. (1962), 'The analysis of proximities: Multidimensional scaling with an unknown distance function. i.', *Psychometrika* **27**(2), 125–140.

Sievert, C. (2020), *Interactive Web-Based Data Visualization with R, plotly, and shiny*, Chapman and Hall/CRC. https://plotly-r.com

Silva, V. & Tenenbaum, J. (2002), 'Global Versus Local Methods in Nonlinear Dimensionality Reduction', *Advances in Neural Information Processing Systems* **15**, 721–728. https://proceedings.neurips.cc/paper_files/paper/2002/file/5d6646aad9bcc0be55b2c82f69750387-Paper.pdf.

Spearman, C. (1961), 'The proof and measurement of association between two things.'.

Tenenbaum, J. B., de Silva, V. & Langford, J. C. (2000), 'A global geometric framework for nonlinear dimensionality reduction', *Science* **290**(5500), 2319–2323. https://www.science.org/doi/abs/10.1126/science.290.5500.2319

Ushey, K., Allaire, J. & Tang, Y. (2024), *reticulate: Interface to Python*. R package version 1.38.0. https://CRAN.R-project.org/package=reticulate.

van der Maaten, L. (2009), Learning a parametric embedding by preserving local structure, *in* 'Artificial Intelligence and Statistics', PMLR, pp. 384–391.

Wang, Y., Huang, H., Rudin, C. & Shaposhnik, Y. (2021), 'Understanding How Dimension Reduction Tools Work: An Empirical Approach to Deciphering t-SNE, UMAP, TriMap, and PaCMAP for Data Visualization', *Journal of Machine Learning Research* **22**(201), 1–73. http://jmlr.org/papers/v22/20-1061.html.

Wickham, H. (2023), *conflicted: An Alternative Conflict Resolution Strategy*. R package version 1.2.0. https://CRAN.R-project.org/package=conflicted.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. & Yutani, H. (2019), 'Welcome to the Tidyverse', *Journal of Open Source Software* **4**(43), 1686. https://doi.org/10.21105/joss.01686.

Wickham, H., Cook, D. & Hofmann, H. (2015), 'Visualizing Statistical Models: Removing the Blindfold', *Statistical Analysis and Data Mining: The ASA Data Science Journal* **8**(4), 203–225. https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.11271.

Wickham, H., Cook, D., Hofmann, H. & Buja, A. (2011), 'tourr: An R Package For Exploring Multivariate Data With Projections', *Journal of Statistical Software* **40**(2), 1–18. http://www.jstatsoft.org/v40/i02/.

Zeileis, A., Fisher, J. C., Hornik, K., Ihaka, R., McWhite, C. D., Murrell, P., Stauffer, R. & Wilke, C. O. (2020), 'colorspace: A Toolbox for Manipulating and Assessing Colors and Palettes', *Journal of Statistical Software* **96**(1), 1–49. https://www.jstatsoft.org/index.php/jss/article/view/v096i01.

Zhu, H. (2024), *kableExtra: Construct Complex Table with kable and Pipe Syntax*. R package version 1.4.0. https://CRAN.R-project.org/package=kableExtra.